

Git Branching and Merging HOL Report

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Step 1: Create a new branch

```
git branch GitNewBranch
```

```
PS G:\My Drive\JAVA-FSE> git branch GitNewBranch
```

Step 2: List branches

```
git branch -a
```

```
PS G:\My Drive\JAVA-FSE> git branch -a
GitNewBranch
* main
remotes/origin/main
```

Step 3: Switch to new branch

```
git checkout GitNewBranch
```

```
PS G:\My Drive\JAVA-FSE> git checkout GitNewBranch
Switched to branch 'GitNewBranch'
```

Step 4: Create a sample file

```
echo "Hello Git" > sample.txt
```

```
PS G:\My Drive\JAVA-FSE> echo "Hello Git" > sample.txt
```

Step 5: Commit changes

```
PS G:\My Drive\JAVA-FSE> git add .
PS G:\My Drive\JAVA-FSE> git commit -m "Add sample file in GitNewBranch"
On branch GitNewBranch
nothing to commit, working tree clean
```

Step 6: Check status

```
git status
```

```
PS G:\My Drive\JAVA-FSE> git status
On branch GitNewBranch
nothing to commit, working tree clean
```

Step 7: Switch to master/main

git checkout main # or git checkout master

```
PS G:\My Drive\JAVA-FSE> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
```

Step 8: Compare branches (CLI)

git diff main GitNewBranch

```
PS G:\My Drive\JAVA-FSE> git diff main GitNewBranch
PS G:\My Drive\JAVA-FSE>
```

Step 9: Visual diff (P4Merge)

git difftool main GitNewBranch

```
PS G:\My Drive\JAVA-FSE> git difftool main GitNewBranch
```

Step 10: Merge branch

git merge GitNewBranch

```
PS G:\My Drive\JAVA-FSE> git merge GitNewBranch
Already up to date.
```

Step 11: View log

git log --oneline --graph --decorate

```

PS G:\My Drive\JAVA-FSE> git log --oneline --graph --decorate
* 9102cee (HEAD -> main, origin/main, GitNewBranch) Add sample file in GitNewBranch
* 6e77c7d Add Git Ignore HOL report
* 988896c Implement .gitignore for log files
* f67aea7 Added Git HOL Commands and Report PDF
* f3fc4fb Remove Git HOL Commands and Report PDF
* 826fcc5 Add Week 6 Git Ignore hands-on exercises
* f25e3e5 Add Week 5 React hands-on exercises
* c3fbd07 Add README.md for Week 5 exercises and objectives
* 4891bb7 Add Week 5 React files
* 9fe31b2 Add Week 4 Microservices with Spring Boot 3 and Spring Cloud
* f1d6712 Add source directory README
* 4a7c0b7 Add Week 3 Spring Core, Maven, Spring Boot and Hibernate modules
* 3240c51 Update Week 1 README
* 23ca687 Add Week 2 exercises and documentation
* 5e5002e Add Week 1 exercises and documentation
* f66c817 Add README for Java Full Stack Engineer program
* 1fba707 Enhance project documentation and update README
* 4860e9c Update Singleton Pattern
* 3848120 Add Week 1 Design Pattern implementations
* 08bda1e Refactor exercises, reorganize folders, and update PL/SQL and DSA files
* 63be4f1 Add gitignore
* bd68d50 Add PL SQL exercises
* d9cd130 Add DSA exercises
PS G:\My Drive\JAVA-FSE>

```

Step 12: Delete merged branch

```
git branch -d GitNewBranch
```

```

PS G:\My Drive\JAVA-FSE> git branch -d GitNewBranch
Deleted branch GitNewBranch (was 9102cee).
PS G:\My Drive\JAVA-FSE>

```

Step 13: Push changes

```
git push origin main
```

```

PS G:\My Drive\JAVA-FSE> git push origin main
Everything up-to-date
PS G:\My Drive\JAVA-FSE>

```